

Deep Learning for Content Moderation on the Shareish Solidarity Platform

Progress Report — Follow-up to April 13, 2026

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May 2026

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Since April 13 — What Changed

This deck covers work completed between April 13 and May 5, 2026. It assumes familiarity with Phase 1 and the two-tier architecture. Thesis chapter drafts are attached separately.

April 13, 2026 — Open question	May 5, 2026 — Status
Phase 2 — LoRA fine-tuning: Reddit-FR adapter pending; joint training hypothesised	Complete. All adapters evaluated; SG-2b × Reddit-FR LoRA is best (F1 0.662)
Phase 3 — Data collection: planned but not started	In progress. Track A (1,500 synthetic items) done; Track B scrapers partially running
Phase 4 — Two-tier architecture: design proposed	Under evaluation. Group 2 done; Tier 1 variant 2c selected; success criterion not yet met
Thesis writing — Early chapter drafts	Advancing. 1,317 lines across all chapters; Chs 2, 3, 5 most active

Phase 2 — LoRA Fine-Tuning: Complete

April 13: Reddit-FR adapter still pending; SG-2b SLURM timeout; joint training hypothesised. All resolved.

Model × Dataset	Baseline F1	LoRA F1	ΔF1	ΔTPR	ΔTNR	Character
SG-2b × Reddit-FR	0.335	0.662	+0.327	+0.415	−0.177	Recall-driven
LG-1B × FHS	0.371	0.557	+0.186	−0.039	+0.324	Precision-driven
SG-2b × FHS	0.413	0.534	+0.121	−0.007	+0.149	Precision-driven

FHS adapters are precision-driven (TNR ↑, models learn what is *not* harmful); Reddit-FR adapter is recall-driven (TPR ↑). This distinction guides Tier 2 selection.

SG-2b × Reddit-FR LoRA selected as Tier 2 — best overall F1 (0.662); recall substantially improved (+0.415 TPR).

Phase 3 — Data Collection: Where We Stand

Track A — Synthetic data (complete)

Generated via Mistral-7B-Instruct; HateCheck-FR functionalities: slurs, leet-speak, character deletion, implicit derogation, counter-speech.

- 1,500 synthetic items; **complete**
- Already used in Phase 4 Group 2 (variant 2c)
- Note: list-prefix artifact ("1. ") stripped before training

Track B will train a content-type classifier (solidarity_exchange vs commercial_listing). No classifier trained yet — awaiting donnerie.be and 2ememain.be.

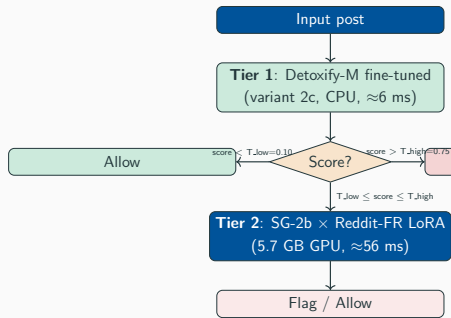
Action Item

Shareish listing data: access or a sample is needed to complete Track B.

Track B — Scrapers

Source	Class	Target	Coll.	Status
donnons.org	solidarity	3,400	3,238	Done
donnerie.be	solidarity	1,000	0	Rerunning (href bug fixed)
2ememain.be	commercial	800	51	Scraper fix needed
Shareish	solidarity	TBD	TBD	Awaiting supervisor access

Phase 4 — Two-Tier Architecture: Current Configuration



Key numbers:

- Tier 1 (variant 2c): Reddit-FR F1 = 0.668, HC-FR F1 = 0.816, FNR = 25.2%
- Tier 2 alone: F1 = 0.640 (511-sample honest holdout)
- Combined (75.5% deferral): F1 = 0.643, avg 47.7 ms
- Speed at 10% deferral: 12.6 ms (4× faster than Tier 2 alone)

Success Criterion

Combined F1 > 0.640 at < 30% deferral — **NOT YET MET**

Phase 4 — Tier 1 Fine-Tuning: Group 2 Results

Four Tier 1 variants evaluated on honest 511-sample holdout (no contamination from earlier Tier 1 training runs).

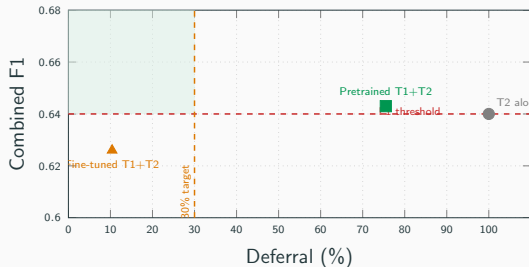
Variant	Reddit-FR F1	HC-FR F1	T1_FNR	Notes
Track B baseline (3 ep)	0.662	—	41.7%	Bimodal score collapse; data leakage detected
2a — 10 epochs	0.615	—	28.5%	Overfits after epoch 3; no net gain
2b — soft labels ($\epsilon=0.05$)	0.619	—	28.3%	T1_FPR: 23.8% $\rightarrow$ 13.5%
2c — synthetic data (ep 2)	0.668	0.816	25.2%	Best: T_high breaks to 0.80

Variant 2c selected as Tier 1 for Group 3. Best checkpoint is epoch 2 — all subsequent runs use `--epochs 2`.

Group 3 Hypothesis

Specialising Tier 2 on the deferred distribution from variant 2c should reduce combined-system FNR and push end-to-end F1 above the success threshold at <30% deferral.

Phase 4 — End-to-End: Speed vs. F1 Trade-off



Config	F1	Deferral	Avg ms
T2 alone	0.640	100%	55.9
Pretrained T1+T2	0.643	75.5%	47.7
Fine-tuned T1+T2	0.626	10.4%	12.6

Speed advantage is real (4× at low deferral), but no configuration simultaneously achieves $F1 > 0.640$ AND deferral $< 30\%$. Group 3 targets this gap.

Remaining Work — Phases 3 & 4

Phase 3 — remaining:

- Rerun donnerie.be scraper (href bug fixed); target 1,000 items
- Fix 2ememain.be Dutch-language filter; target 800 items
- Train content-type classifier once Track B complete
- Evaluate classifier on Shareish listing sample

Definition of Done

Architecture complete when: combined F1 > 0.640 AND deferral < 30%.

Phase 4 — remaining:

- Simulate end-to-end with variant 2c + SG-2b LoRA on 511-sample holdout
- Group 3: Tier 2 specialisation on deferred distribution at $T=(0.10, 0.75)$
- Stretch: SG-2b retrain ($LR=10^{-4}$, balanced joint adapter 1:1 FHS:Reddit-FR)

Thesis Writing — Current Status

Chapter	Topic	Lines	Status
Introduction	Motivation, Shareish, research plan	76	Draft
Ch 1 — Background	NLP, hate speech, content moderation	193	Draft
Ch 2 — Framework	Models, datasets, metrics	227	In progress
Ch 3 — Phase 1 Results	Baseline evaluation	351	In progress
Ch 4 — Data Strategy	Scraping + synthetic generation	71	Stub
Ch 5 — LoRA Fine-tuning	Phase 2 results	184	In progress
Ch 6 — Conclusions	Architecture recommendation	117	Draft
Total		1,317	—

Chapters 3 and 5 are most advanced, reflecting completed experimental phases. Chapter 4 will expand once Phase 3 data results are available.

Attached to this email: [list chapters]. **Feedback most welcome on:** [priority sections].

Questions & Feedback Sought

1. **Phase 4 success criterion** — The current target ($F1 > 0.640$ at $< 30\%$ deferral) was self-defined. Is this the right criterion, or should it be reframed (e.g. $FNR < 25\%$, or a latency target alone)?
2. **Group 3 scope** — Given the combined F1 gap is only 0.014, is Tier 2 specialisation worth the remaining SLURM time, or should effort shift to writing?
3. **Content-type classifier (Track B)** — Is this a thesis-depth contribution or better placed in an appendix?
4. **Chapter 4 — Data Strategy** is currently a 71-line stub. Guidance on expected scope and depth is welcome.
5. **Attachments** — Which of the attached chapter drafts should be prioritised for review?